Lecture-10
Inviscid Flows I: Euler Equation

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Learning Objectives

By the end of this lecture, you should be able to

1. Justify the **ideal fluid approximation** from a **fluid dynamic perspective**.
2. Establish the **fundamental difference** between the physical forces present in an **inviscid versus** a **viscous** flow field.
3. Derive the vector form of the **Euler equation.**
4. Explain the concept of **convective acceleration** and how that makes the Euler equation **nonlinear.**

Introduction to Euler equations

- Up to this point in our kinematics lectures, we have analyzed how fluid elements move, rotate, stretch, and dilate. However, we never asked **what forces** were responsible for the fluid element's motion.
- In a real fluid, forces arise from **gravity**, **pressure**, and an **internal friction**, which we call viscosity.
- For many **vital engineering applications**, such as high-speed airflow over an airplane wing or water rushing through a massive dam spillway, the **internal inertial forces completely overwhelm the viscous forces**.
- The **governing equation** for these frictionless fluid flows is referred to as the **Euler equation**.

Euler's Equation

$$\rho\frac{D\vec{V}}{Dt}=\rho\vec{g}-\nabla p$$

$$\rho\left(\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+\upsilon\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}\right)=\rho g_x-\frac{\partial p}{\partial x}$$

$$\rho\left(\frac{\partial \upsilon}{\partial t}+u\frac{\partial \upsilon}{\partial x}+\upsilon\frac{\partial \upsilon}{\partial y}+w\frac{\partial \upsilon}{\partial z}\right)=\rho g_y-\frac{\partial p}{\partial y}$$

$$\rho\left(\frac{\partial w}{\partial t}+u\frac{\partial w}{\partial x}+\upsilon\frac{\partial w}{\partial y}+w\frac{\partial w}{\partial z}\right)=\rho g_z-\frac{\partial p}{\partial z}$$

$$\rho\left(\frac{\partial V_r}{\partial t}+V_r\frac{\partial V_r}{\partial r}+\frac{V_\theta}{r}\frac{\partial V_r}{\partial \theta}+V_z\frac{\partial V_r}{\partial z}-\frac{V_\theta^2}{r}\right)=\rho g_r-\frac{\partial p}{\partial r}$$

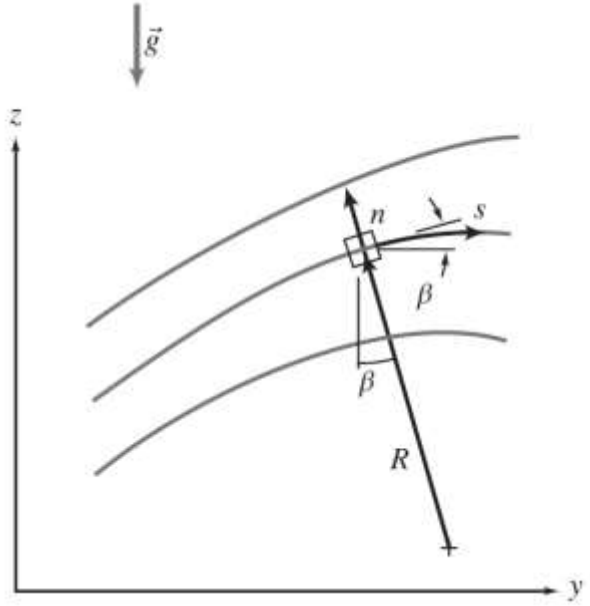
$$\rho\left(\frac{\partial V_\theta}{\partial t}+V_r\frac{\partial V_\theta}{\partial r}+\frac{V_\theta}{r}\frac{\partial V_\theta}{\partial \theta}+V_z\frac{\partial V_\theta}{\partial z}+\frac{V_rV_\theta}{r}\right)=\rho g_\theta-\frac{1}{r}\frac{\partial p}{\partial \theta}$$

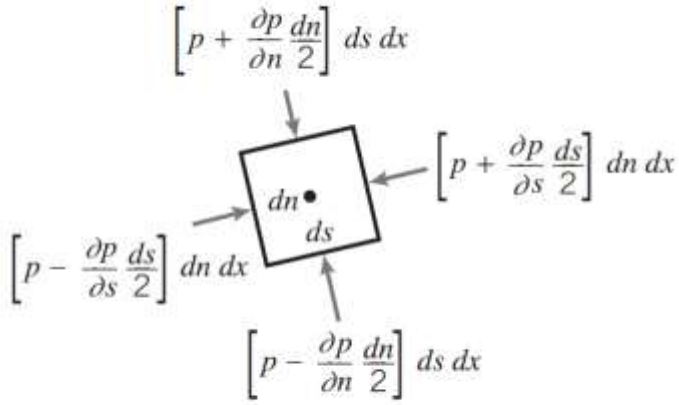
$$\rho\left(\frac{\partial V_z}{\partial t}+V_r\frac{\partial V_z}{\partial r}+\frac{V_\theta}{r}\frac{\partial V_z}{\partial \theta}+V_z\frac{\partial V_z}{\partial z}\right)=\rho g_z-\frac{\partial p}{\partial z}$$

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Euler's Equation along streamline coordinates

- For many practical scenarios, it is useful to write the Euler equations along **streamline coordinates**





$$\frac{1}{\rho}\frac{\partial p}{\partial s} = -V\frac{\partial V}{\partial s}$$

$$\frac{1}{\rho}\frac{\partial p}{\partial n} = \frac{V^2}{R}$$

Revisiting the learning objectives

1. Justify the **ideal fluid approximation** from a **fluid dynamic perspective**.
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